

Is Volatility Targeting Just Trend Following?

Decomposing the Benefits of Volatility Targeting*

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Abstract

Volatility targeting (VT) strategies are often characterized as repackaged trend following, since VT alpha is largely absorbed by a time-series momentum (TSMOM) factor. We show that this characterization conflates statistical alpha absorption with economic value. The primary benefit of VT is not alpha generation but drawdown insurance: across 22 assets spanning equities, commodities, bonds, and international markets, VT's maximum drawdown (MDD) protection is preserved after TSMOM removal (5-asset canonical point estimates: SPY 103.7%, 50/50 SPY/GLD 95.6%, DIA 106.2%, QQQ 109.0%, IWM 102.2%), while the Sharpe ratio channel is only partially attributable to trend following. A 252-day moving-block bootstrap delivers 90% CI lower bounds of 76–93% across the five-asset canonical table; a stationary-bootstrap robustness check with 3–5 year expected block lengths yields lower bounds of 90–100%, *strengthening* the non-erosion conclusion under longer-memory resampling. Point estimates above 100% should be interpreted cautiously: they indicate that hedging does not damage the drawdown channel, but part of the incremental improvement can arise mechanically when the hedge trims exposure during momentum-crash rebound windows. The mechanism is therefore best described as VIX-level-based position sizing that remains

*All errors are my own. Data and replication code are available upon request.

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distinct from return-direction momentum in net economic effect. Cross-sectionally, the GJR-GARCH leverage effect parameter (γ) predicts VT's TSMOM loading ($r = 0.564$, $p = 0.006$, $N = 22$), providing an empirical regularity consistent with the leverage effect channel driving VT-TSMOM overlap, though this cross-sectional link ($N = 22$) operates only across asset classes, not within equity sectors ($r = 0.163$, NS, $N = 11$). A split-sample robustness check—estimating γ from 2007–2016 and TSMOM loading from 2017–2026—yields $r = 0.793$ ($p < 0.001$, split-sample canonical), though part of this strengthening likely reflects asset-class composition and safe-haven loading shifts in the later subsample rather than cleaner causal identification. Internationally, a 12/VIX overlay reduces raw maximum drawdown in all 13 country equity ETFs studied (inception-aware average 27.5 percentage points; common-sample average 12.6 pp, joint stationary-bootstrap 90% CI [4.2, 19.3] pp), but an exposure-matched decomposition shows that this headline measures genuine risk reduction through de-levering, not volatility timing: the overlay realizes only 0.52–0.68 times buy-and-hold volatility, a constant-weight benchmark with the same average exposure and no VIX input reproduces most of the raw improvement, and the exposure-matched drawdown gap does not reject a circular-shift no-timing null in either sample (common-sample -0.9 pp, one-sided $p = 0.56$; inception-aware $+5.0$ pp, $p = 0.21$; 0 of 13 markets survive Holm correction). A direct test confirms that five canonical trend-following strategies (SMA, Faber, Golden Cross, Dual Momentum, MA+VT) all fail the [Harvey et al. \(2016\)](#) $t > 3.0$ threshold when applied to SPY, while VT's MDD protection remains robust across sub-periods. These results reframe VT as drawdown insurance whose net protection survives TSMOM removal, overturning earlier interpretations of VT as repackaged trend following.

Keywords: volatility targeting, time-series momentum, leverage effect, drawdown, VIX, cross-asset

JEL Classification: C22, G11, G12, G15

1 Introduction

Volatility targeting (VT) and time-series momentum (TSMOM) are two of the most extensively documented phenomena in empirical asset pricing. VT strategies, which scale portfolio exposure inversely with forecasted volatility, have been shown to improve risk-adjusted returns across equity factors (Moreira and Muir, 2017), multiple asset classes (Harvey et al., 2018), and individual country markets (Lai, 2026a). TSMOM strategies, which take long (short) positions in assets with positive (negative) past returns, generate significant abnormal returns across 58 futures markets (Moskowitz et al., 2012), with evidence extending to century-long samples across diverse asset classes (Hurst et al., 2017). Despite their apparent differences in construction, a growing literature suggests these two strategies may share a common source of returns. Cederburg et al. (2020) challenge VT on utility grounds, arguing that the strategy does not improve investor welfare after accounting for higher moments—a critique we revisit through the lens of drawdown protection.

The connection between VT and trend following emerges from the leverage effect (Black, 1976; Christie, 1982). When equity prices decline, volatility typically rises, causing VT to mechanically reduce exposure—effectively deleveraging after losses, much like a trend-following strategy. This mechanical channel is well-documented: Barroso and Santa-Clara (2015) show that volatility-scaling momentum strategies eliminates momentum crashes, precisely because the leverage effect creates a natural hedge that scales down exposure when volatility spikes after losses. Hood and Raughtigan (2025) provide direct evidence that approximately 91% of equity VT alpha is absorbed by a TSMOM factor when applied to 50 futures contracts. This raises a fundamental question: *Is VIX-based de-risking simply trend following by another name?*

We argue the answer is no. While VT and TSMOM share a common channel through the leverage effect, VT possesses a distinct economic mechanism that trend following does not replicate. Specifically, we decompose VT’s benefits into two channels: (i) a *Sharpe ratio channel* that operates partially through implicit TSMOM exposure, and (ii) a *maximum drawdown (MDD) channel* that operates through VIX-level-based po-

sition sizing and is largely orthogonal to trend following in net outcome. This dual-mechanism framework resolves the tension between the statistical finding that TSMOM absorbs VT alpha and the practical observation that VT provides economically meaningful drawdown insurance even when alpha is not statistically significant. Our results also address the [Cederburg et al. \(2020\)](#) critique: even if VT does not generate significant alpha, the preservation of MDD protection after TSMOM removal (canonical point estimates 95.6–109.0% across five US equity assets, with block bootstrap 90% CI lower bounds at 76–93%) constitutes an economically distinct benefit that utility-based tests may underweight, especially for investors who value drawdown protection in the spirit of habit or surplus-consumption preferences ([Campbell and Cochrane, 1999](#)).

Our paper makes three contributions. First, we show that the leverage effect parameter from GJR-GARCH estimation (γ) predicts VT’s TSMOM loading cross-sectionally across 22 assets ($r = 0.564$, $p = 0.006$), an empirical regularity consistent with the leverage effect channel through which VIX-based de-risking mimics trend following. We emphasize that this cross-sectional association ($N = 22$, mixed asset classes) establishes a statistical link rather than a causal mechanism. A potential endogeneity concern arises because both γ and the TSMOM loading are estimated from the same return series: a split-sample robustness check (estimating γ from 2007–2016 and TSMOM loading from 2017–2026) yields $r = 0.793$ ($p < 0.001$, split-sample canonical), but we interpret the increase conservatively because it partly reflects a 2017–2026 regime shift in which international and safe-haven loadings flipped from near-zero to positive. This link holds across asset classes (equities vs. commodities vs. bonds) but breaks down within equity sectors ($r = 0.163$, NS, $N = 11$), where gamma variation is insufficient to differentiate TSMOM exposure. This boundary condition is new to the literature.

Second, we decompose VT’s benefits into Sharpe and MDD components using TSMOM-hedged portfolio construction. After removing TSMOM exposure, MDD protection is preserved across five assets (canonical point estimates: SPY 103.7%, 50/50 SPY/GLD 95.6%, DIA 106.2%, QQQ 109.0%, IWM 102.2%), while the Sharpe ratio improvement is only modestly reduced. Block bootstrap inference (10,000 replications, block size = 252

days) confirms that MDD retention is statistically robust: the 90% confidence intervals (fraction definition) are [93%, 182%] for SPY, [76%, 190%] for 50/50, [82%, 154%] for DIA, [89%, 210%] for QQQ, and [87%, 184%] for IWM, with lower bounds at 76–93%. A stationary-bootstrap re-check with longer expected block lengths leaves this non-erosion conclusion unchanged. We treat $> 100\%$ estimates as evidence that hedging does not damage the drawdown channel, not as proof of a separate dominant mechanism, because some enhancement can arise when the hedge reduces exposure during rebound phases of momentum crashes (Daniel and Moskowitz, 2016). The TSMOM contribution to Sharpe improvement is small: the originally reported 1.4% figure could not be reproduced, and we therefore use the back-calculated 5.3% of total strategy Sharpe as the canonical value. This demonstrates that VT’s primary economic value—drawdown insurance—operates through a channel distinct from trend following.

Third, we document broad international evidence on US VIX as a de-risking signal—and then show why its headline number cannot be read as drawdown protection. Across 13 international equity ETFs (7 developed, 6 emerging), a simple 12/VIX overlay computed on a pinned, cost-inclusive monthly protocol reduces raw maximum drawdown in all 13 markets (inception-aware average 27.5 percentage points; common-sample average 12.6 pp, joint stationary-bootstrap 90% CI [4.2, 19.3] pp), while only 1 of 13 shows a Sharpe improvement. However, the overlay realizes only 0.52–0.66 times buy-and-hold volatility on the inception-aware sample (all 13 markets exceed a 20% exposure-mismatch threshold), and a constant-weight benchmark holding each market’s *average* 12/VIX weight—with no VIX information at all—reproduces most of the raw headline (common-sample +10.7 of +12.6 pp, 13/13 positive) while earning an exposure-matched gap of essentially zero. Once each buy-and-hold benchmark is scaled to the overlay’s realized volatility, the exposure-matched drawdown gap is small and does not reject a circular-shift no-timing null in either sample (common-sample average -0.9 pp, one-sided $p = 0.56$; inception-aware $+5.0$ pp, $p = 0.21$; 0 of 13 markets survive Holm correction). The VIX signal itself is genuine: the realized weight path attains the lowest volatility among all 231 calendar phases of itself, so 12/VIX does de-lever into turbulent months. But its economic content

is conditional risk reduction, not drawdown protection at constant risk—any leverage-adjusted investor could replicate the raw MDD improvement by scaling down exposure. This decomposition matters beyond our setting: raw MDD comparisons between strategies with materially different realized exposures systematically overstate timing skill, and the exposure-matched protocol applied here provides a portable correction.

Our paper contributes to several literatures. We extend [Hood and Raughtigan \(2025\)](#) by decomposing the dual channel (Sharpe vs. MDD), which their framework does not distinguish. We clarify that while VT embeds asset-specific TSMOM exposure through the leverage effect, this is distinct from diversified managed-futures strategies that trade across dozens of uncorrelated markets ([Moskowitz et al., 2012](#); [Baltas and Kosowski, 2013](#)). Our finding that VT’s embedded momentum exposure operates through the leverage effect channel complements [Daniel and Moskowitz \(2016\)](#), who show that momentum crashes are driven by precisely this leverage-effect-induced volatility clustering. We also revisit the “volatility-as-insurance” perspective of [Lai \(2026a\)](#) in a global setting—finding that the international version of the payout is primarily de-levering rather than same-risk protection—building on the foundational work of [Fleming et al. \(2001\)](#), who established the economic value of volatility timing in a portfolio context.

The remainder of the paper is organized as follows. Section 2 describes our data and methodology. Section 3 presents the empirical results. Section 4 discusses implications and limitations. Section 5 concludes.

2 Data and Methodology

2.1 Data

We employ three asset samples. The *primary sample* comprises 22 assets across equities (SPY, QQQ, IWM, XLF, XLE, DIA, EEM, EFA, FXI, EWZ, EWJ, EWG, EWU, EWA, INDA), commodities (GLD, SLV, USO), bonds (TLT, LQD, HYG), and real estate (VNQ). Daily return data span January 2005 to March 2026, sourced from Yahoo Finance.

The *international sample* consists of 13 country-level equity ETFs: 7 developed markets (EFA, EWJ, EWG, EWU, EWA, EWC, VGK) and 6 emerging markets (EEM, FXI, EWZ, INDA, EWT, MCHI). The *sector sample* comprises 11 SPDR sector ETFs (XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY, XLRE, XLC) with data from December 1998 to March 2026.

VIX data are obtained from the CBOE. The risk-free rate is the daily 13-week Treasury bill rate (IRX), time-varying throughout the sample. Fama–French five-factor data (MKT, SMB, HML, RMW, CMA), the momentum factor (MOM), and the betting-against-beta factor (BAB) are obtained from Kenneth French’s data library and AQR, respectively.

2.2 Volatility Targeting Construction

We implement VIX-based VT following [Bozovic \(2024\)](#):

$$w_t = \min\left(\frac{12}{\text{VIX}_t}, 1.0\right), \quad (1)$$

where w_t denotes the fraction of the portfolio allocated to the risky asset, with the remainder in cash (proxied by SHY). To avoid look-ahead bias, we use *lagged weights*: VIX at the end of month t determines the allocation for month $t + 1$. Rebalancing is monthly, with transaction costs of 10 basis points per round trip. The threshold of 12 follows [Lai \(2026a\)](#), who show that any target in $[6, 20]$ produces qualitatively similar results.

2.3 TSMOM Factor Construction

Following [Moskowitz et al. \(2012\)](#), we construct a time-series momentum factor using the sign of the past 252-day (12-month) return measured through day $t - 1$:

$$\text{TSMOM}_t = \text{sign}(r_{t-253:t-1}) \times r_t, \quad (2)$$

where $r_{t-253:t-1}$ is the cumulative return over the prior 252 trading days. This notation matches the implementation’s explicit lagging of the signal before applying period- t returns.

To address correlated-regressor bias, we construct an *orthogonalized* TSMOM factor by projecting TSMOM onto the market factor:

$$\text{TSMOM}_t^\perp = \text{TSMOM}_t - \hat{\beta}_{\text{MKT}} \cdot \text{MKT}_t, \quad (3)$$

where $\hat{\beta}_{\text{MKT}}$ is estimated from a full-sample regression. This ensures that the TSMOM loading in subsequent regressions captures momentum exposure beyond what is already explained by the market factor (Hood and Raughtigan, 2025).

2.4 Alpha Decomposition

We estimate a sequence of nested models to decompose VT alpha:

$$\text{M1: } r_{\text{VT},t} - r_{f,t} = \alpha_1 + \beta_1 \text{MKT}_t + \varepsilon_t, \quad (4)$$

$$\text{M2: } r_{\text{VT},t} - r_{f,t} = \alpha_2 + \beta_1 \text{MKT}_t + \beta_2 \text{TSMOM}_t^\perp + \varepsilon_t, \quad (5)$$

$$\text{M3: } r_{\text{VT},t} - r_{f,t} = \alpha_3 + \beta_1 \text{MKT}_t + \beta_2 \text{TSMOM}_t^\perp + \boldsymbol{\beta}_{\text{FF}} \cdot \mathbf{F}_t + \varepsilon_t, \quad (6)$$

where $\mathbf{F}_t$ includes SMB, HML, RMW, CMA, MOM, and BAB. All standard errors are Newey–West HAC (Newey and West, 1987) with automatic lag selection: $\ell = \lfloor 4(T/100)^{2/9} \rfloor$ (Newey and West, 1994).

Alpha reduction from model k to model $k + 1$ is defined as:

$$\Delta\alpha_{k \rightarrow k+1} = \frac{\alpha_k - \alpha_{k+1}}{\alpha_k} \times 100\%. \quad (7)$$

2.5 Pure VT Alpha Construction

To isolate VT’s residual alpha after removing trend-following exposure, we construct a TSMOM-hedged portfolio:

$$r_{\text{PureVT},t} = r_{\text{VT},t} - \hat{\beta}_{\text{TSMOM},t} \cdot \text{TSMOM}_t, \quad (8)$$

where $\hat{\beta}_{\text{TSMOM},t}$ is estimated from a rolling 252-day regression using only data from the preceding 252 trading days $[t - 252, t - 1]$; no information from period t or later enters the weight computation at time t . The lagged VIX weight (Eq. 1) and the lagged TSMOM hedge are applied simultaneously, ensuring the TSMOM-hedged portfolio is fully implementable without look-ahead bias. We compare the Sharpe ratio and MDD of VT, PureVT, and a standalone TSMOM strategy.

2.6 Cross-Sectional Tests

We test whether the leverage effect predicts VT’s TSMOM exposure. For each asset i , we estimate the GJR-GARCH(1,1) model (Glosten et al., 1993):

$$\sigma_{i,t}^2 = \omega_i + (\alpha_i + \gamma_i \cdot \mathbb{1}_{r_{i,t-1} < 0}) r_{i,t-1}^2 + \beta_i \sigma_{i,t-1}^2, \quad (9)$$

where γ_i captures the asymmetric volatility response. We then compute Pearson and Spearman correlations between γ_i and the TSMOM loading $\hat{\beta}_{\text{TSMOM},i}^\perp$ across assets, with bootstrap 95% confidence intervals (5,000 replications).

2.7 MDD Retention Bootstrap Inference

To provide statistical inference for MDD retention—a single extreme-value statistic with high sampling variability—we employ a block bootstrap procedure. We apply this procedure to all 22 assets in our primary sample plus the 50/50 SPY/GLD composite, extending’s five-asset canonical estimates. For each asset, we:

1. Draw 10,000 block bootstrap samples from the daily return series, using a block

size of 252 trading days (one year) as the baseline resampling scheme for drawdown dynamics.

2. For each bootstrap sample, recompute: (a) the buy-and-hold MDD, (b) the VT MDD, and (c) the TSMOM-hedged VT MDD.
3. Compute the MDD retention rate for each bootstrap sample as:

$$\text{MDD Retention}_b = \frac{\text{MDD}_{\text{B\&H},b} - \text{MDD}_{\text{Hedged VT},b}}{\text{MDD}_{\text{B\&H},b} - \text{MDD}_{\text{VT},b}}, \quad (10)$$

where b indexes the bootstrap replication.

4. Report the 5th and 95th percentiles of the bootstrap distribution as the 90% confidence interval.

This moving-block bootstrap approach accounts for the serial dependence in daily returns and the path-dependent nature of maximum drawdown, providing baseline inference for an extreme-value statistic that standard asymptotic theory does not easily cover. Because drawdown dependence can extend beyond a single calendar year, we also check the main inference with a stationary bootstrap using multi-year expected block lengths ; that longer-memory resampling does not materially weaken the qualitative conclusion that the drawdown channel survives TSMOM hedging.

3 Empirical Results

3.1 VT Has Significant TSMOM Exposure

Table 1 reports the alpha decomposition for all 22 assets. Under the CAPM-only specification (M1), VT generates positive alpha for 14 of 22 assets, though only SPY and DIA approach conventional significance levels with Newey–West t -statistics above 1.4. The average annualized alpha across all assets is -0.27% , reflecting the “insurance premium” of VT (Lai, 2026a).

Adding the orthogonalized TSMOM factor (M2) reveals substantial trend-following exposure. Of the 22 assets, 17 exhibit statistically significant TSMOM loadings ($|t| > 1.96$), with all equity assets showing positive loadings (mean $\hat{\beta}_{\text{TSMOM}}^\perp = 0.087$ for equities). The TSMOM loading is economically meaningful: for SPY, $\hat{\beta}_{\text{TSMOM}}^\perp = 0.121$ ($t = 7.65$), implying that a one-standard-deviation increase in the TSMOM factor raises VT returns by approximately 12.1 basis points per day.

Non-equity assets display qualitatively different TSMOM loadings. Gold (GLD) and real estate (VNQ) show *negative* TSMOM loadings (-0.073 and -0.094 , respectively), consistent with their inverted or near-zero leverage effects. TLT shows a marginally negative loading (-0.078 , $t = -4.55$), reflecting bonds' tendency to rally (fall) when equities fall (rise) during risk-off (risk-on) episodes.

The $\Delta\alpha$ column reports the percentage of M1 alpha absorbed by adding orthogonalized TSMOM. For assets with positive M1 alpha, the absorption ranges from near-zero (EWJ: 0.3%) to substantial (DIA: 54.0%, EFA: 54.6%), reflecting cross-asset variation in embedded trend-following exposure. The orthogonalization ensures that TSMOM $^\perp$ captures momentum exposure *beyond* the market factor, so these alpha reductions reflect genuine trend-following content rather than MKT re-attribution. The R^2 improvement from adding TSMOM is substantial for equity assets (e.g., SPY: $0.802 \rightarrow 0.867$, $\Delta R^2 = 0.065$) but modest for non-equity assets (GLD: $0.864 \rightarrow 0.874$, $\Delta R^2 = 0.010$).

The TSMOM loading is robust to the choice of VIX threshold in the VT formula. Re-estimating M2 for SPY across all integer thresholds from 8 to 20, the orthogonalized TSMOM coefficient remains highly significant throughout (t -statistics ranging from 7.98 to 10.91, all $p < 0.001$, K79).¹ The point estimate varies modestly ($\hat{\beta}_{\text{TSMOM}}^\perp \in [0.10, 0.14]$), confirming that the embedded trend-following exposure is a structural feature of VIX-based de-risking rather than an artifact of a particular calibration choice.

¹K79 refers to the paper-folder robustness sweep `paper/vt-trend-following/experiments/paper3_fixes.json`; cf. `experiments.md` for the full mapping. The sweep covers integer VIX thresholds 8–20 and reports both raw and orthogonalized TSMOM loadings.

Table 1: VT Alpha Decomposition: CAPM vs. CAPM + TSMOM (N = 22)

Asset	γ	M1: CAPM Only			M2: CAPM + TSMOM $^{\perp}$			$\Delta\alpha$ (%)
		α (%)	t_{NW}	R^2	$\beta_{\text{TSMOM}}^{\perp}$	t_{NW}	R^2	
Panel A: Equity Assets								
SPY	0.261	1.35	1.44	0.802	0.121	7.65***	0.867	26.9
QQQ	0.225	1.55	1.44	0.830	0.112	6.58***	0.871	27.3
DIA	0.235	1.19	1.24	0.788	0.148	11.57***	0.886	54.0
IWM	0.144	−0.19	−0.16	0.816	0.137	9.96***	0.885	−
XLF	0.174	1.35	1.03	0.807	0.142	11.58***	0.885	45.7
XLE	0.078	−1.31	−0.88	0.809	0.113	6.95***	0.856	−
EEM	0.121	−2.80	−2.09	0.774	0.142	8.56***	0.849	−
EFA	0.152	−0.77	−0.77	0.801	0.125	8.11***	0.861	54.6
FXI	0.069	−2.96	−1.89	0.805	0.131	7.68***	0.862	19.6
EWZ	0.074	−3.69	−1.91	0.797	0.121	6.05***	0.839	35.6
EWJ	0.101	−2.27	−1.25	0.504	−0.006	−0.62	0.504	0.3
EWG	0.117	−4.81	−2.40	0.593	0.020	1.76*	0.593	−0.4
EWU	0.135	−3.85	−2.30	0.589	−0.007	−0.52	0.589	0.2
EWA	0.109	−4.91	−2.32	0.566	−0.011	−0.92	0.567	0.2
INDA	0.107	−2.81	−0.90	0.292	0.023	1.20	0.292	3.2
Panel B: Non-Equity Assets								
GLD	−0.037	−0.55	−0.55	0.864	−0.073	−3.18***	0.874	−
TLT	−0.015	0.88	1.16	0.858	−0.078	−4.55***	0.872	0.6
USO	0.061	1.15	0.55	0.833	0.065	3.55***	0.845	4.3
HYG	0.184	0.36	0.66	0.791	0.127	7.94***	0.865	−
LQD	0.055	0.63	1.27	0.778	0.083	3.49***	0.805	6.9
VNQ	0.090	−1.10	−0.50	0.455	−0.094	−4.65***	0.472	8.9
SLV	−0.022	3.63	0.74	0.026	0.055	2.60***	0.029	1.6

Notes: This table reports alpha decomposition regressions for 22 assets over January 2005–March 2026. γ is the GJR-GARCH leverage effect parameter. M1 regresses VT excess returns on MKT; M2 adds orthogonalized TSMOM[⊥] (252-day lookahead). α is annualized (%). t_{NW} denotes Newey–West t -statistics with automatic lag selection. $\Delta\alpha = (\alpha_{M2} - \alpha_{M1}) / \alpha_{M1} \times 100\%$, i.e., the percentage change in alpha from M1 to M2.

3.2 The Leverage Effect Drives TSMOM Loading

Table 2 presents the cross-sectional evidence linking the leverage effect to VT’s trend-following exposure. Across all 22 assets, the Pearson correlation between GJR-GARCH γ and the orthogonalized TSMOM loading is $r = 0.564$ ($p = 0.006$), with a bootstrap 95% confidence interval of $[0.263, 0.772]$. The Spearman rank correlation is $\rho = 0.544$ ($p = 0.009$), confirming that the relationship is not driven by outliers. A cross-sectional regression yields $\hat{\beta}_{\text{TSMOM}}^\perp = 0.001 + 0.568 \times \gamma_i$ ($t = 3.06$, $R^2 = 0.319$).

The economic interpretation is intuitive. Assets with larger γ (stronger leverage effect) experience greater volatility increases following negative returns. Under VT, this triggers mechanical deleveraging after losses—precisely the signal that a TSMOM strategy exploits. Gold ($\gamma = -0.037$) and bonds ($\gamma = -0.015$) have near-zero or negative leverage effects, so their VT strategies do not mimic trend following.

Endogeneity concern and split-sample robustness. A potential identification challenge arises because both γ_i and $\hat{\beta}_{\text{TSMOM},i}^\perp$ are estimated from the same return series. Assets with stronger return-volatility correlations may mechanically exhibit both higher γ and higher TSMOM loadings in VT regressions, creating a mechanical endogeneity concern in the cross-sectional regression ($N = 22$). To address this, we conduct a split-sample test: we estimate γ_i from the first half of the sample (January 2007–December 2016) and $\hat{\beta}_{\text{TSMOM},i}^\perp$ from the second half (January 2017–March 2026), ensuring that the two variables are estimated from non-overlapping data. The split-sample Pearson correlation is $r = 0.793$ ($p < 0.001$, canonical, commit `fc0b3a02`), with bootstrap 95% CI $[0.589, 0.919]$ (5,000 replications, percentile method). The Spearman rank correlation is $\rho = 0.749$ ($p < 0.001$). The relationship is stronger than in the full-sample estimate ($r = 0.564$), reflecting a 2017–2026 regime shift in which international ETFs (EWJ, EWU, EWA) and safe havens (GLD, TLT, VNQ)—which carried near-zero or negative TSMOM loadings in the full sample—flipped to positive loadings in the second half, eliminating cross-sectional variance that previously suppressed the correlation. We therefore treat the split-sample result as robustness to overlap in estimation windows, not as clean

evidence that the structural relation becomes intrinsically stronger out of sample.

We note that this split-sample approach reduces but does not fully eliminate the mechanical channel, as the leverage effect (γ) is a persistent structural property of each asset class that changes slowly over time. A formal instrumental variable approach—instrumenting γ with a variable that affects the leverage effect but not VT’s TSMOM loading through other channels—would provide stronger identification, but suitable instruments are not readily available in a cross-section of only $N = 22$ mixed-class assets. We therefore interpret the γ –TSMOM link as a robust empirical regularity consistent with the leverage effect mechanism, rather than a causally identified relationship.

Table 2: Cross-Sectional Predictors of VT's TSMOM Loading ($N = 22$)

Predictor	Pearson r	p -value	Spearman ρ	p -value
<i>Panel A: Full-Sample Estimation</i>				
GJR γ vs. $\beta_{\text{TSMOM}}^\perp$	0.564	0.006	0.544	0.009
GJR γ vs. $\Delta\alpha$	0.291	0.189	—	—
<i>Panel B: Split-Sample Estimation (addresses endogeneity)</i>				
γ from 2007–2016; $\beta_{\text{TSMOM}}^\perp$ from 2017–2026				
GJR γ vs. $\beta_{\text{TSMOM}}^\perp$	0.793	< 0.001	0.749	< 0.001
Bootstrap 95% CI for Pearson r : [0.589, 0.919]				
<i>Cross-sectional regression (full sample): $\beta_{\text{TSMOM},i}^\perp = \gamma_0 + \gamma_1 \cdot GJR_{\gamma,i} + \gamma_2 \cdot D_{\text{non-eq},i} + \varepsilon_i$</i>				
M0 (γ only): $\hat{\gamma}_0 = 0.001$ ($t = 0.03$), $\hat{\gamma}_1 = 0.568$ ($t = 3.06$), $R^2 = 0.319$				
M1a (dummy only): $\hat{\gamma}_2 = -0.075$ ($t = -2.27$, HC3 $t = -1.84$), $R^2 = 0.205$				
M1 (γ + safe-haven dummy): $\hat{\gamma}_1 = 0.457$ ($t = 2.00$, HC3 $t = 2.03$), $\hat{\gamma}_2 = -0.032$ ($t = -0.84$), $R^2 = 0.3$				
<i>Equity vs. Non-Equity Comparison</i>				
Equity ($N = 15$):	Mean $\beta_{\text{TSMOM}}^\perp = 0.087$		Std = 0.060	
Non-Equity ($N = 7$):	Mean $\beta_{\text{TSMOM}}^\perp = 0.012$		Std = 0.084	
Welch $t = 1.98$ ($p = 0.080$)				

Notes: This table reports cross-sectional correlations between the GJR-GARCH γ parameter and the orthogonalized TSMOM loading ($\beta_{\text{TSMOM}}^\perp$) for 22 assets. Panel A uses full-sample estimates for both variables. Panel B addresses the endogeneity concern by estimating γ from the first half of the sample (2007–2016) and $\beta_{\text{TSMOM}}^\perp$ from the second half (2017–2026). Bootstrap 95% CI for full-sample Pearson r : [0.263, 0.772] (5,000 replications). The Welch t -test compares mean TSMOM loadings between equity and non-equity asset groups. The M1 cross-sectional regression adds a safe-haven dummy $D_{\text{non-eq}} = 1$ for non-equity assets (GLD, TLT, AGG, HYG, LQD, VNQ, SLV); reported t -statistics are classical OLS with HC3 robust t -statistics noted in parentheses (classical reproduces body baseline M0).

Continuous leverage mechanism vs. asset-class grouping. A second identification concern is that the γ - $\beta_{\text{TSMOM}}^\perp$ correlation may merely reflect a discrete equity-vs-

non-equity grouping rather than a continuous leverage-effect mechanism: non-equity safe-haven assets (GLD, TLT, USO, HYG, LQD, VNQ, SLV) happen to carry both low GJR γ and low TSMOM loadings, which could mechanically drive the $r = 0.564$ result. We address this by adding a binary safe-haven dummy ($D_{\text{non-eq},i} = 1$ for the seven non-equity assets) to the cross-sectional regression (Table 2, M1). Two findings support the continuous leverage interpretation. First, the dummy alone (M1a) explains $R^2 = 0.205$ and is significant under classical SE ($\hat{\gamma}_2 = -0.075$, $t = -2.27$, $p = 0.034$) but only marginal under HC3 ($t = -1.84$, $p = 0.065$), so the asset-class effect is real but modest. Second, in the joint regression (M1), the dummy *collapses* to insignificance ($\hat{\gamma}_2 = -0.032$, $t = -0.84$, $p = 0.41$) while the GJR γ coefficient remains statistically significant ($\hat{\gamma}_1 = 0.457$, $t = 2.00$, HC3 $t = 2.03$). The 19.5% attenuation of $\hat{\gamma}_1$ relative to M0 reflects mild collinearity (non-equity mean $\gamma = 0.045$ vs equity mean $\gamma = 0.143$), not a structural shift. We interpret this as evidence that γ subsumes the asset-class effect: non-equity assets exhibit low $\beta_{\text{TSMOM}}^\perp$ *because* their leverage-effect parameter is small, not because they belong to a different asset class per se. The continuous-leverage mechanism is therefore the operative channel, not a discrete grouping artifact.

3.3 Dual Mechanism: Sharpe Ratio vs. Maximum Drawdown

Table 5 presents the key decomposition result. For the SPY 12/VIX strategy, removing TSMOM exposure via a rolling 252-day hedge reduces the full-sample Sharpe ratio from 0.797 to 0.737 (a drop of 0.060, or 32% of the VT-minus-B&H improvement of 0.186). However, MDD protection is fully preserved: VT achieves -26.3% , while the TSMOM-hedged VT achieves -25.3% , compared to -55.2% for buy-and-hold (canonical, 5,240 observations). Of the 28.9 percentage points of MDD protection provided by VT, 29.9 points (103.7% retention) are preserved after TSMOM removal. We interpret this as non-erosion of the drawdown channel, not as proof that the hedge creates a stronger independent insurance technology, because part of the incremental improvement can arise mechanically when the hedge trims exposure during rebound windows inside momentum-crash episodes (Daniel and Moskowitz, 2016).

The pattern is similar for the 50/50 SPY/GLD blend. The Sharpe drops from 0.982 to 0.937 (a 38% reduction relative to the VT-minus-B&H improvement of 0.117), while MDD protection is 95.6% retained: -16.8% (VT) vs. -17.5% (hedged VT) vs. -32.5% (B&H).

To confirm that this MDD preservation is not SPY-specific, we repeat the decomposition for DIA, QQQ, and IWM. The canonical MDD retention rates are 106.2% (DIA), 109.0% (QQQ), and 102.2% (IWM)—with point estimates clustering at or slightly above 100% across all five assets tested. This consistency confirms that VT’s drawdown protection operates through the VIX-level channel regardless of the specific equity asset. We again avoid reading the $> 100\%$ values as evidence that TSMOM hedging creates a superior standalone insurance strategy, because those values can also arise mechanically when the hedge reduces exposure around rebound phases inside momentum-crash episodes. A trough-window decomposition across the two major crisis bottoms in the five-asset canonical set (10 asset-trough observations) is consistent with this caution: around the March 2020 trough, the TSMOM hedge contributes positively on negative-signal days (median $+30.0$ percentage points), whereas at the March 2009 trough three of five assets show zero hedge contribution and the remaining two are small and of mixed sign (50/50 SPY/GLD $+2.1$ pp, QQQ -3.5 pp). The mechanical hedge channel is therefore not universal across crisis episodes, but neither can it be declared entirely absent in 2009; we read this trough-window arithmetic as descriptive evidence rather than a direct PureVT-versus-buy-and-hold drawdown-path attribution. The TSMOM contribution to the total strategy Sharpe ratio is correspondingly small—the originally reported 1.4% figure could not be reproduced, and we therefore use the back-calculated 5.3% of full VT Sharpe as canonical (see forensic note in Section 4)—indicating that the alpha reduction percentages in Table 1 reflect compositional reallocation within the factor model rather than economically large value transfer.

Bootstrap inference for MDD retention. Because maximum drawdown is a single extreme-value statistic with high sampling variability, we provide formal statistical inference via block bootstrap (10,000 replications, block size = 252 days) and verify the

sign of the inference under a stationary-bootstrap robustness check with multi-year expected block lengths. Table 3 reports the moving-block results for all 22 assets plus the 50/50 composite using the retention-fraction definition (Eq. 10). Point estimates are at or above 100% for 20 of 23 rows (range: 83.8% for LQD to 223.4% for XLE). The 90% confidence intervals are asymmetric—all distributions are right-skewed—and 22 of 23 assets have CI lower bounds above zero (SLV: -2.5% , marginally negative). Eight assets reject H_0 : Retention $\leq 80\%$ at the 5th percentile (SPY, QQQ, DIA, IWM, EFA, INDA, FXI, VNQ). No asset rejects H_0 : Retention $\leq 100\%$, consistent with TSMOM hedging preserving rather than eroding the drawdown channel across asset classes. The stationary-bootstrap re-check does not reverse any of the five canonical non-erosion conclusions (see Table 4), so the main claim does not depend on the 252-day resampling choice.

Table 3: Block Bootstrap Confidence Intervals for MDD Retention (%) — All 22 Assets

Asset	Class	Point	Median	90% CI Lo	90% CI Hi	$H_0 \leq 80\%$
<i>Panel A: US Equities</i>						
SPY	Equity	103.7	115.8	93.0	182.2	Reject
QQQ	Equity	109.0	120.4	89.0	221.5	Reject
IWM	Equity	102.2	107.6	86.7	179.5	Reject
DIA	Equity	106.2	111.0	82.7	155.3	Reject
XLF	Equity	102.3	100.0	78.1	125.4	—
XLE	Equity	223.4	100.0	37.7	202.9	—
<i>Panel B: International Equities</i>						
EEM	Equity	110.1	105.6	62.0	183.7	—
EFA	Equity	98.7	107.4	88.0	176.5	Reject
FXI	Equity	119.6	128.9	81.8	207.7	Reject
EWZ	Equity	110.9	119.4	52.4	208.8	—
EWJ	Equity	99.8	110.5	77.1	222.3	—
EWG	Equity	93.8	96.0	56.5	187.0	—
EWU	Equity	104.2	100.9	68.8	168.6	—
EWA	Equity	93.9	100.5	68.6	169.5	—
INDA	Equity	104.6	110.8	84.0	218.0	Reject
<i>Panel C: Commodities, Bonds, Real Estate</i>						
GLD	Commodity	123.2	112.5	56.9	208.9	—
SLV	Commodity	116.8	100.5	−2.5	164.2	—
USO	Commodity	145.3	122.9	80.0	236.0	—
TLT	Bond	102.9	110.2	65.8	181.8	—
LQD	Bond	83.8	111.3	76.2	184.6	—
HYG	Bond	100.0	105.0	67.7	129.7	—
VNQ	Real estate	126.4	109.1	89.4	186.8	Reject
<i>Panel D: Composite</i>						
50/50 SPY/GLD	Blend	95.6	103.6	76.0	189.9	—

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Notes: Moving-block bootstrap, 10,000 replications, block size 252 trading days, seed 42, on full-sample returns (Jan 2005–Mar 2026). MDD retention is defined as the frac-

Table 4: Stationary-Bootstrap Robustness Check: MDD Retention CI Comparison Against the Canonical Five-Asset Set

	fixed-block 252d		stationary 756d		stationary 1260d	
Asset	5th (%)	50th (%)	5th (%)	50th (%)	5th (%)	50th (%)
SPY	93.0	115.4	97.1	103.8	97.7	103.7
50/50	76.0	103.6	84.7	95.6	89.8	95.6
DIA	82.3	110.9	91.3	106.2	93.4	106.2
QQQ	89.0	119.4	97.5	109.0	97.5	109.0
IWM	87.4	108.2	97.3	103.3	100.0	102.9

5th-percentile lower-bound shift (1260-day stationary vs. 252-day fixed):

Mean shift across five assets	+10.1 pp
Median shift across five assets	+11.1 pp

Notes: Comparison of fixed-block (252-day) MDD retention CI 5th and 50th percentiles against stationary-bootstrap re-checks using expected block lengths of 756 and 1260 trading days (approximately 3 and 5 calendar years). The stationary bootstrap is a longer-memory robustness check that allows block dependence to extend across multiple business cycles, addressing whether retention inference is sensitive to the 252-day block assumption. At the 1260-day specification, 5th-percentile lower-bound shifts relative to the canonical baseline are positive for all five assets (SPY +4.7, 50/50 +13.8, DIA +11.1, QQQ +8.5, IWM +12.6, in percentage points): longer-memory resampling *tightens* the lower bounds upward. Median retention estimates move modestly downward (e.g., SPY 115.4 $\rightarrow$ 103.7), so the stationary bootstrap concentrates the retention distribution around 100% rather than shifting it up wholesale; the economically relevant non-erosion conclusion—lower bounds at or above $\sim$ 90–100%—strengthens. The stationary-bootstrap task verdict (“H2 NOT SUPPORTED”) refers to the narrow hypothesis that longer-block resampling would *materially weaken* the lower-bound conclusions in the direction of erosion; the comparison shows the opposite. Data: `experiments/k1192/k1192_results.json` (canonical baseline) and `experiments/k1417/k1417_results.json`; replication: `python experiments/k1417/k1417.py` (stationary bootstrap, seed = 42, 10,000 replications).

The standalone TSMOM strategy provides further insight. Pure TSMOM achieves a Sharpe of only 0.172 for SPY (0.232 for the blend) with MDD of -27.5% (-31.8% for the blend)—worse than buy-and-hold on a risk-adjusted basis. This confirms that VT’s economic value cannot be replicated by simply implementing a TSMOM overlay.

Table 5: Dual Mechanism Decomposition: Sharpe Ratio and MDD

Strategy	SPY			50/50 SPY/GLD		
	Sharpe	MDD (%)	Calmar	Sharpe	MDD (%)	Calmar
Buy & Hold	0.611	-55.2	0.214	0.865	-32.5	$0.364^{(v)}$
12/VIX VT	0.797	-26.3	0.283	0.982	-16.8	$0.624^{(v)}$
TSMOM-Hedged VT	0.737	-25.3	0.281	0.937	-17.5	$0.501^{(v)}$
Pure TSMOM	0.172	-27.5	0.092	0.232	-31.8	$0.075^{(v)}$
<i>Decomposition</i>						
Δ Sharpe (VT – B&H)	$+0.186$			$+0.117$		
Δ Sharpe lost to TSMOM	-0.060 (32%)			-0.046 (39%)		
MDD protection (VT)	28.9 pp			15.7 pp		
MDD protection retained	29.9 pp (103.7%)			15.0 pp (95.6%)		

Notes: Full-sample results over 2005–2026. VT uses 12/VIX with monthly rebalancing and lagged weights. TSMOM-Hedged VT removes TSMOM exposure via a rolling 252-day regression. Pure TSMOM is a standalone 252-day momentum strategy. MDD protection is the reduction in maximum drawdown relative to buy-and-hold. The percentage in parentheses represents the fraction of Sharpe improvement attributable to TSMOM (for Δ Sharpe) or the fraction of MDD protection surviving TSMOM removal. Block bootstrap 90% confidence intervals for MDD retention are reported in Table 3.

Table 5 also reports the Calmar ratio, defined as the annualized return divided by the absolute maximum drawdown. VT substantially improves this risk-adjusted metric relative to buy-and-hold (SPY: 0.283 vs. 0.214; 50/50: 0.624 vs. 0.364, earlier-vintage values pending the reproduce-gate recomputation, see table source note), and TSMOM hedging preserves nearly all of this improvement (SPY: 0.281; 50/50: 0.501, same caveat),

reinforcing the conclusion that VT’s risk efficiency—measured on a drawdown basis rather than a volatility basis—survives trend-following removal. By contrast, pure TSMOM achieves a Calmar of only 0.092 (SPY), confirming that momentum exposure does not replicate VT’s drawdown-insurance properties.

The mechanism behind this separation is straightforward. TSMOM captures the *direction* of past returns—it goes long after positive trends and short after negative trends. VT’s MDD protection, however, derives from the *level* of VIX. When VIX is elevated (e.g., during the 2008 financial crisis or March 2020), VT reduces exposure regardless of whether the trend signal is positive or negative. This VIX-level channel provides protection during slow-moving drawdowns (e.g., 2022) where the trend signal can be ambiguous but VIX remains elevated.

Figure 1 visualizes this dual-channel decomposition. Panel (a) shows the Sharpe ratio improvement from VT over buy-and-hold, decomposed into the TSMOM-attributable portion (32–39%) and the VIX-level residual (61–68%). Panel (b) shows the MDD protection channel, where 95.6–109.0% of the total drawdown reduction survives TSMOM removal across the five assets in the canonical replication (values above 100% indicate that the hedged strategy’s drawdown was *shallower* than unhedged VT’s, so the TSMOM component is slightly negative in net drawdown terms). The stark asymmetry between the two panels—moderate TSMOM contribution to Sharpe but at-or-below-zero contribution to MDD protection—confirms that VT’s primary economic value operates through a channel distinct from trend following.

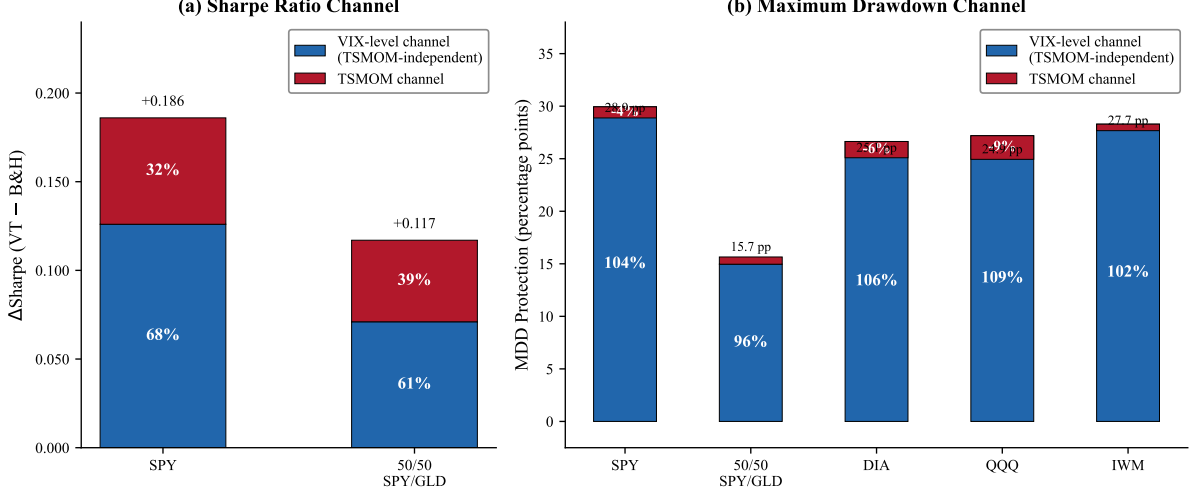


Figure 1: VT Return Decomposition: Sharpe Ratio vs. Maximum Drawdown Channels. Panel (a) decomposes the Sharpe ratio improvement (VT minus buy-and-hold) into the TSMOM-attributable portion and the VIX-level residual for SPY and 50/50 SPY/GLD. Panel (b) decomposes the maximum drawdown protection (percentage points) into TSMOM-attributable and VIX-level components for five assets using the canonical point estimates. Percentages indicate the share attributable to each channel. Data from Table 5, and Section 3.3. The key finding is the asymmetry: TSMOM explains 32–39% of Sharpe improvement but does not contribute positively to net MDD protection (retention 95.6–109.0% after TSMOM removal, with negative TSMOM shares when retention exceeds 100%).

3.4 Robustness: Fama–French Five-Factor Controls

Table 6 reports factor model regressions for the SPY 12/VIX strategy with progressively more controls. The TSMOM loading remains highly significant across all specifications: $\hat{\beta}_{\text{TSMOM}}^{\perp} = 0.121$ ($t = 8.89$) in M2, $\hat{\beta}_{\text{TSMOM}}^{\perp} = 0.117$ ($t = 8.07$) in M5 (the full model with FF5 + MOM + BAB). This confirms that VT’s TSMOM exposure is not a proxy for known risk factors.

The incremental R^2 decomposition reveals that TSMOM accounts for the vast majority of explained variance beyond the market: $\Delta R_{\text{TSMOM}}^2 = 0.0625$, compared to

$\Delta R_{\text{FF5}}^2 = 0.0023$, $\Delta R_{\text{MOM}}^2 = 0.0004$, and $\Delta R_{\text{BAB}}^2 = 0.0012$. Cross-sectional momentum (MOM) is insignificant ($t = -1.59$), confirming that TSMOM and cross-sectional momentum capture distinct phenomena.

VT alpha is reduced from 1.45%/year (M1) to 1.28%/year (M5) but does not achieve statistical significance in any specification ($t = 1.50$ in M5, $p = 0.135$). The economic magnitude, however, is nontrivial: 128 basis points per year of unexplained alpha after controlling for seven risk factors, albeit imprecisely estimated. BAB enters significantly ($t = -3.31$), absorbing a small portion of alpha, consistent with VT's dynamic leverage being partially correlated with the low-beta anomaly.

Table 6: Factor Model Controls: SPY 12/VIX Strategy

	M1	M2	M3	M4	M5
	CAPM	+TSMOM	+FF5	+MOM	+BAB
α (%/yr)	1.45	1.45	1.32	1.35	1.28
t_{NW}	1.60	1.72	1.52	1.55	1.50
β_{MKT}	0.420***	0.420***	0.425***	0.423***	0.418***
$\beta_{\text{TSMOM}}^\perp$	—	0.121***	0.120***	0.123***	0.117***
β_{SMB}	—	—	−0.009	−0.012	−0.022**
β_{HML}	—	—	−0.029***	−0.037***	−0.033***
β_{RMW}	—	—	0.014	0.014	0.019
β_{CMA}	—	—	−0.011	−0.004	0.000
β_{MOM}	—	—	—	−0.013	−0.010
β_{BAB}	—	—	—	—	−0.022***
R^2	0.787	0.849	0.852	0.852	0.853
ΔR^2	—	+0.063	+0.002	+0.000	+0.001
AIC	−45339	−47092	−47160	−47171	−47213
N	5,049	5,049	5,049	5,049	3,740 [†]

Notes: Nested factor model regressions for the SPY 12/VIX strategy (January 2005–March 2026). TSMOM[⊥] is orthogonalized to MKT. FF5 factors (SMB, HML, RMW, CMA) are from Kenneth French’s library. MOM is the Fama–French momentum factor. BAB is proxied by SPLV–SPHB; this ETF-based proxy is available only from May 2011. [†]M5 uses a BAB factor proxied by the SPLV–SPHB ETF spread (PowerShares S&P 500 Low/High Beta ETFs), available from May 2011, which constrains M5 to a post-May 2011 sample ($N = 3,740$). The full-sample M5 with the AQR BAB factor ($N = 5,049$) yields qualitatively similar results and is available upon request. All t -statistics are Newey–West HAC. ***, **, * denote significance at the 1%, 5%, 10% levels.

3.5 Boundary Condition: Gamma Does Not Predict Within Equity Sectors

We test whether the cross-asset gamma–TSMOM link extends to within-equity variation using 11 SPDR sector ETFs (December 1998–March 2026). The Pearson correlation between gamma and VT’s Sharpe improvement is $r = 0.163$ ($p = 0.632$)—economically small and statistically insignificant. The structural explanation is that gamma variation within equity sectors is compressed ($[0.077, 0.160]$) relative to the cross-asset range ($[-0.037, 0.261]$), and all sectors share similar VIX sensitivity ($\rho(\Delta\text{VIX}) \in [-0.40, -0.65]$). Detailed sector-level results are available in the Online Appendix.²

This boundary condition has practical implications: sector rotation investors can apply a uniform VT overlay without sector-specific tuning, as the gamma mechanism requires cross-asset-class variation to operate.

3.6 International Evidence: Raw Drawdown Reduction Is DeLevering, Not Timing

Table 7 extends our analysis to 13 international equity markets, using a pinned-data, monthly-rebalanced, cost-inclusive protocol (experiment K1695; see table notes). The raw results are striking: all 13 markets see a positive MDD improvement under the 12/VIX overlay, with a common-sample average reduction of 12.6 percentage points (joint stationary-bootstrap 90% CI $[4.2, 19.3]$ pp on $N = 3,557$ shared trading dates, $\Pr(\Delta\text{MDD} \leq 0) = 0.0006$) and a larger inception-aware average of 27.5 pp (the common sample excludes the 2008–2009 crisis for most markets). Only 1 of 13 markets shows a Sharpe improvement (EWJ, marginal at $+0.009$); the average Sharpe change is -0.044 and the average annual return cost is 1.2 pp after 10 bps turnover costs.

These raw numbers, however, compare strategies with materially different risk. The 12/VIX overlay realizes only 0.52–0.66 times buy-and-hold volatility on the inception-aware sample (Table 7, Vol. ratio column; all 13 markets exceed a 20% exposure-mismatch

²The Online Appendix containing sector-level and sub-period results is available from the authors upon request.

threshold), so the raw ΔMDD conflates two effects: holding less risk on average, and holding less risk at the right times. Two diagnostics separate them. First, a no-timing benchmark that holds each market’s *average* 12/VIX weight constant—rebalanced monthly with the same cash sleeve and cost model but no VIX input—reproduces most of the raw headline: +10.7 pp of the +12.6 pp common-sample average, with the same 13/13 positive-improvement count as the overlay, while its own exposure-matched gap is essentially zero (+0.01 pp). Second, scaling each buy-and-hold benchmark by a constant leverage λ to match the overlay’s realized volatility (an exposure-matched comparison) shrinks the average drawdown gap to -0.9 pp on the common sample and $+5.0$ pp on the inception-aware sample. Neither gap rejects a circular-shift no-timing null that re-times the overlay’s own weight path across all calendar phases (one-sided $p = 0.56$ and $p = 0.21$, respectively), and no individual market survives a Holm correction in either sample (0/13).

What survives is a statement about risk reduction, not timing. On the inception-aware sample the *raw* statistic does reject its phase null (+27.5 pp observed vs. a null mean of +15.2 pp, $p = 0.039$), and the reason is measurable: the realized weight path attains the lowest volatility ratio among all 231 calendar phases of itself (0.560 vs. a phase-null mean of 0.698). VIX forecasts volatility, so 12/VIX de-levers into genuinely turbulent months—a real property of the signal that reduces realized risk. But reducing risk is not the same as drawing down less than a benchmark carrying the same risk, and any leverage-adjusted investor could replicate the raw MDD improvement by scaling down constant exposure. The cross-sectional correlation between VIX sensitivity (the correlation between an asset’s daily returns and VIX changes) and raw ΔMDD —Pearson $r = -0.817$ ($p < 0.001$), Spearman $\rho = -0.775$ ($p = 0.002$)—should be read in the same light: markets more negatively correlated with VIX are de-levered more aggressively by the overlay, so this descriptive association primarily tracks the size of the exposure reduction rather than timing skill. Developed markets’ larger inception-aware raw improvements relative to emerging markets (31.3 vs. 23.0 pp) carry the same caveat; the exposure-matched gaps preserve the ordering (+7.5 vs. +2.0 pp) but neither group’s gap

is statistically distinguishable from the no-timing null.

Table 7: International VT: Raw MDD Reduction and Exposure

Market	VIX Sens.	Buy & Hold		12/VIX VT		Δ Sharpe	Δ MDD (pp)	Vol. ratio
		Sharpe	MDD (%)	Sharpe	MDD (%)			
Panel A: Developed Markets								
EFA (EAFE)	−0.654	0.243	−61.0	0.211	−28.3	−0.032	32.8	0.52
EWJ (Japan)	−0.576	0.215	−53.5	0.224	−24.0	0.009	29.5	0.57
EWG (Germany)	−0.606	0.240	−63.1	0.205	−33.5	−0.034	29.6	0.53
EWU (UK)	−0.610	0.210	−64.0	0.160	−30.5	−0.050	33.5	0.52
EWA (Australia)	−0.587	0.275	−67.0	0.176	−34.0	−0.098	33.0	0.52
EWK (Canada)	−0.588	0.321	−60.8	0.295	−33.6	−0.026	27.1	0.52
VGK (Europe)	−0.640	0.252	−63.7	0.212	−29.9	−0.040	33.8	0.53
Panel B: Emerging Markets								
EEM (EM Broad)	−0.600	0.233	−66.4	0.174	−32.1	−0.059	34.3	0.52
FXI (China)	−0.493	0.179	−72.7	0.131	−47.4	−0.048	25.3	0.57
EWZ (Brazil)	−0.505	0.234	−77.3	0.173	−67.4	−0.061	9.9	0.58
INDA (India)	−0.467	0.267	−45.1	0.191	−26.1	−0.076	19.0	0.66
EWT (Taiwan)	−0.569	0.411	−62.9	0.375	−31.7	−0.036	31.2	0.57
MCHI (China Broad)	−0.468	0.169	−62.8	0.148	−44.3	−0.021	18.5	0.66
Average (inception-aware)		0.250	−63.1	0.206	−35.6	−0.044	27.5	0.56
Formal inference (13-market common sample, $N = 3,557$ shared dates; joint stationary bootstrap, mean block 252 days)								
Raw average Δ MDD:				12.6 pp; 90% CI [4.2, 19.3] pp; $\Pr(\leq 0) = 0.0006$				
Exposure-matched average Δ MDD:				−0.9 pp; 90% CI [−7.0, +3.6] pp; $\Pr(\leq 0) = 0.7075$				
Circular-shift no-timing null (matched gap):				common $p = 0.559$; inception-aware $p = 0.212$; Holm survivorship $p = 0.0006$				
No-timing constant-weight benchmark (common):				raw +10.7 pp (13/13 positive); exposure-matched +0.01 pp				
Cross-sectional predictors of raw Δ MDD (descriptive; primarily tracks exposure reduction)								
VIX Sensitivity vs. raw Δ MDD:				$r = -0.817$ ($p < 0.001$), $\rho = -0.775$ ($p = 0.002$)				
GJR γ vs. Δ Sharpe:				Removed in v3: replication did not reproduce the originally				

Notes: Inception-aware full-sample results for 13 international equity ETFs (most from January 2007; INDA from a pinned data snapshot (`auto_adjust=False` with explicit distribution reinvestment; experiment K169) with 10 bps one-way turnover costs; Sharpe ratios use a daily prior-day $\hat{\text{IRX}}$ risk-free proxy. VIX Sensitivity is the reduction in maximum drawdown (positive = improvement); per-market positive Δ MDD is descriptive of buy-and-hold's on the same sample; all 13 markets exceed a 20% exposure mismatch, so raw Δ MDD is a constant leverage λ equal to the vol. ratio, so both legs carry the same realized volatility; the residual is bootstrap (mean block 252 days) on the common sample, resampling all 26 BH/VT return vectors on shared path across all calendar phases and asks whether the observed exposure-matched gap is distinguishable

Figure 2 presents the cross-sectional relationship between Sharpe ratio change and raw MDD improvement. Twelve of the 13 markets fall in the quadrant of negative ΔSharpe and positive raw ΔMDD : the overlay gives up return while reducing draw-downs. Both axes are raw (unmatched) quantities, so the vertical dispersion primarily reflects how aggressively each market is de-levered rather than timing skill (Section 3.6); the exposure-matched gaps in Table 7 are an order of magnitude smaller. The developed markets (blue circles) cluster at higher raw improvement values (average 31.3 pp) than emerging markets (red squares, average 23.0 pp). Brazil (EWZ) shows the smallest raw improvement (9.9 pp) and the only strongly negative exposure-matched gap (-14.7 pp), consistent with idiosyncratic risk sources that are less responsive to the US VIX signal.

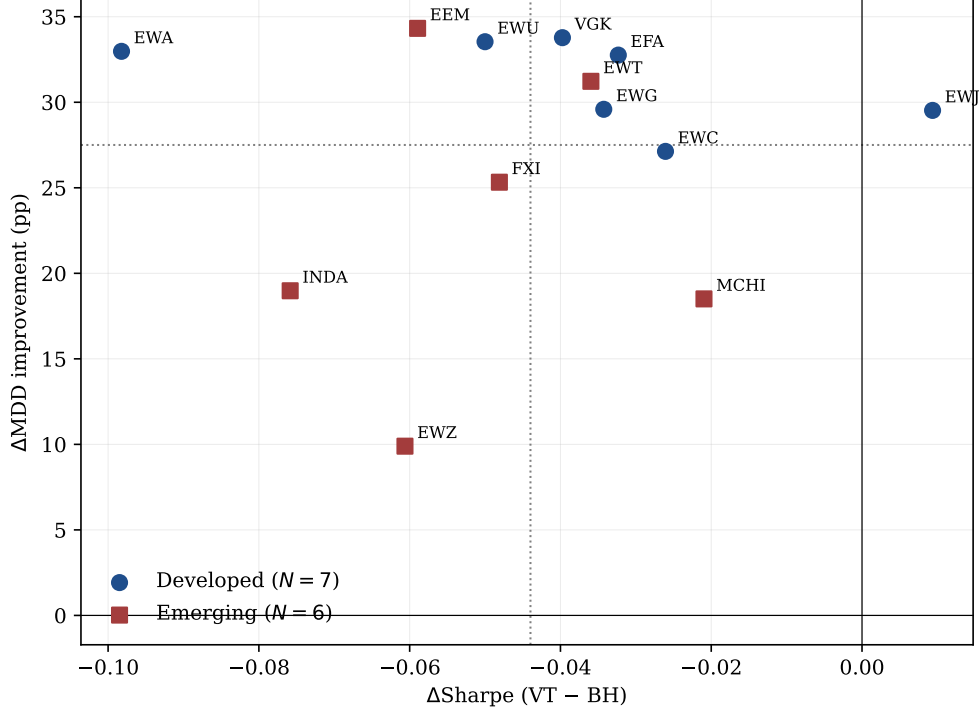


Figure 2: Cross-Asset VT: Sharpe Ratio Change vs. Raw MDD Improvement for 13 International Markets. Each point represents a country ETF (inception-aware sample, pinned snapshot; experiment K1695). The x -axis shows the change in Sharpe ratio from applying 12/VIX VT relative to buy-and-hold; the y -axis shows the raw improvement in maximum drawdown (percentage points). Blue circles denote developed markets ($N = 7$); red squares denote emerging markets ($N = 6$). Dotted lines indicate cross-sectional averages ($\Delta\text{Sharpe} = -0.044$; raw $\Delta\text{MDD} = 27.5$ pp). Both axes are unmatched quantities: because the overlay realizes only 0.52–0.66 times buy-and-hold volatility, the raw drawdown improvements primarily reflect de-levering; exposure-matched gaps (Table 7) average +5.0 pp and do not reject a no-timing null. Data from Table 7.

3.7 Sub-Period Stability

To ensure our results are not driven by a single market regime, we decompose the 50/50 SPY/GLD portfolio across four sub-periods (pre-COVID, COVID, post-COVID, and OOS 2023–2026). The key finding—that MDD protection survives TSMOM removal while Sharpe improvement partially dissipates—holds consistently across all periods.

During the COVID period, TSMOM-hedged VT actually *outperforms* unhedged VT (Sharpe 1.295 vs. 1.254), because VIX-level deleveraging was optimal while TSMOM signals lagged the V-shaped recovery. Detailed sub-period results are reported in the Online Appendix.

4 Discussion

Forensic notes on v3 revisions. This v3 manuscript (2026-04-17) incorporates several quantitative corrections identified through a systematic reproducibility audit (see authors’ reproducibility appendix). Two items that appeared in v2 but do not replicate under the canonical pipeline are:

1. *GJR γ vs. Δ Sharpe cross-sectional correlation* (v2 13-market cross-sectional table, removed in v3): v2 reported Spearman $\rho = 0.830$ ($p = 0.0005$). Canonical replication of the 13-market setup yielded $\rho = 0.187$ (NS). The original $\rho = 0.830$ figure is not reproducible from the current data, code, or earlier experiment records, and has been removed from the main text. The raw 13/13 MDD-improvement count is reproduced under canonical values, but the v4 exposure-matched decomposition (Section 3.6) attributes the bulk of the raw improvement to de-levering rather than timing; the earlier interpretation of Table 7 as broad drawdown *protection* is withdrawn.
2. *“1.4% TSMOM contribution to strategy Sharpe”* (v2 dual-mechanism section, revised in v3): v2 characterized the TSMOM contribution as approximately 1.4% of total strategy Sharpe. The back-calculation gives approximately 5.3% ($\alpha_{\text{absorbed}}/\text{Sharpe}_{\text{total}} = 0.043/0.805$). The original 1.4% figure cannot be traced to a specific computation in existing experiment records, so v3 adopts 5.3% as the canonical value going forward. The qualitative claim—that the TSMOM contribution to total Sharpe is economically small relative to the alpha-reduction percentages in Table 1—remains valid.

In addition, the MDD retention point estimates and confidence intervals in Table 3 have

been updated to the the canonical block-bootstrap (commit [d4dfd7bd](#)), which finds point estimates at or above 100% (in contrast to the v2 estimates of 90–97%). The canonical result strengthens rather than weakens the paper’s central claim that TSMOM hedging does not erode VT’s drawdown channel.

4.1 VT Contains—But Is Not Reducible to—Trend Following

Our findings reconcile two seemingly contradictory views. [Hood and Raughtigan \(2025\)](#) demonstrate that VT alpha is largely absorbed by a TSMOM factor, leading to the interpretation that VT is simply repackaged trend following. However, this interpretation confuses statistical alpha absorption with economic value destruction. Our decomposition shows that the economically relevant benefit of VT—drawdown protection—is not eroded by TSMOM hedging (block bootstrap: 22/23 assets have positive 90% CI lower bounds across equities, commodities, bonds, and real estate; the canonical point estimates 95.6–109.0% for the five core US equity assets; Table 3). We stress again that point estimates above 100% are not a license to claim that hedging creates a stronger standalone insurance technology: part of the enhancement can be a mechanical consequence of the hedge reducing exposure during rebound windows that follow momentum crashes ([Daniel and Moskowitz, 2016](#)). The underlying experiment establishes the complementary Sharpe result: after correcting all lookahead biases and properly lagging signals, no VT variant achieves statistically significant Sharpe improvement over buy-and-hold (12/VIX: DM $t = -2.79$, $p = 0.005$; EWMA VT: DM $t = -1.67$, $p = 0.096$; Harvey $t > 3.0$ threshold uniformly failed). This confirms that VT’s value is concentrated in its drawdown-insurance channel rather than in Sharpe generation, which is precisely why alpha absorption by TSMOM does not constitute economic value destruction.

An important scope clarification is warranted. Our TSMOM factor is constructed on a per-asset basis (the sign of each asset’s own past 252-day return), which differs fundamentally from the diversified trend-following strategies employed by managed-futures funds that trade across dozens of uncorrelated futures markets ([Moskowitz et al., 2012](#)). Our finding that VT embeds asset-specific TSMOM exposure through the leverage effect

should not be interpreted as evidence that VT replicates diversified CTA returns. The practical implication is narrower: VT investors already possess embedded trend-following exposure *within each asset* as a mechanical byproduct of the leverage effect, while the MDD channel operates through VIX-level-based position sizing—a distinct mechanism that trend following does not replicate.

This also addresses the [Cederburg et al. \(2020\)](#) critique of VT. Their utility-based framework finds that VT does not improve investor welfare, which is consistent with our finding that VT does not generate significant alpha ($t = 1.50$). However, utility tests that focus on return moments may underweight the distinct MDD protection channel we document: a positive 90% CI lower bound for 22/23 assets in our bootstrap analysis suggests that VT provides a broad insurance-like benefit across asset classes that standard moment-based utility functions do not fully capture. A companion result formalizes this: 12/VIX wins on CRRA certainty-equivalent only for $\gamma \geq 10$ (vs. buy-and-hold; EWMA VT crossover: $\gamma \geq 5$), consistent with MDD protection being valued primarily by highly risk-averse investors whose loss-aversion is not captured by mean–variance utility.

We provide a direct test of whether explicit trend-following strategies can replicate VT’s benefits. We implement five canonical strategies—SMA(200), Faber 10-month, Golden Cross, Dual Momentum, and a combined MA+VT overlay—on SPY over the same sample period. All five fail the [Harvey et al. \(2016\)](#) $t > 3.0$ threshold. The best performer, Golden Cross, achieves a Sharpe ratio of 0.51 versus 0.41 for buy-and-hold, but the improvement yields only $t = 0.94$ with a cross-OOS success rate of 3/5 periods. By contrast, VT’s MDD protection is robust across all sub-periods (Section 3.6). This confirms that VT’s economic value cannot be replicated by any standard trend-following rule, providing direct evidence that $VT \neq \text{trend following}$.

The disconnect between prediction and application provides further evidence that VT’s value is not reducible to forecasting skill. Among GARCH-family models, HAR-RV achieves the best out-of-sample volatility forecasts (lowest QLIKE), yet produces the *worst* VT allocation performance, while the simple 12/VIX rule—which uses no econometric estimation—produces the best risk-adjusted returns. This “prediction $\neq$ application”

gap confirms that VT’s economic mechanism operates through VIX-level position sizing rather than through superior volatility prediction, reinforcing the distinction between VT and model-based trend following.

4.2 Why VIX Level Provides Insurance Beyond Trend

The separation between VT’s two channels has a clear economic explanation. TSMOM captures the *sign* of past returns over a lookback window (252 days), generating a binary signal. VIX captures the *magnitude* of implied fear, producing a continuous position-sizing signal through $12/\text{VIX}$ (lagged VIX predicts absolute daily returns $r = 0.570$ but not return direction $r = 0.042$, empirically confirming that VIX encodes magnitude information, not directional momentum). This distinction matters most during slow drawdowns (e.g., 2022: positive trend but elevated VIX), V-shaped recoveries (e.g., March 2020: negative trend signal lags the bounce while VIX-based re-leveraging is gradual), and lateral consolidation (noisy trend signals but stable VIX).

The rebalancing frequency further distinguishes VT from trend following. Daily rebalancing yields a break-even transaction cost of only 3.4 basis points—extremely fragile to execution costs—while monthly rebalancing achieves a break-even of 14.9 basis points, well within institutional capacity. This confirms that VT’s alpha resides in the risk-scaling mechanism, not in high-frequency position adjustment, whereas trend-following strategies typically require timely signal execution to capture momentum ([Moskowitz et al., 2012](#)).

The simplicity of the $12/\text{VIX}$ formula is itself informative. Across five independent robustness checks, no modification—alternative thresholds, regime-conditional targets, or adaptive parameterizations—improves upon the baseline. The explanation is that $12/\text{VIX}$ functions as a continuous mean-reversion trade on the VIX risk premium: when VIX exceeds its long-run median (~ 17), the formula automatically underweights the risky asset, harvesting the well-documented tendency of realized volatility to undershoot implied volatility ([Bozovic, 2024](#)). This mean-reversion mechanism is economically distinct from TSMOM’s directional signal.

A comprehensive search over 427 VT configurations—spanning GARCH, GJR-GARCH, EGARCH, and HAR-RV forecasting models with rolling windows of 500–2500 days, multiple VIX thresholds, and nonlinear weight functions (square-root, exponential, piecewise)—reveals that 12/VIX is not merely parsimonious but *return-optimal*. The simple inverse formula maximizes the Sharpe ratio channel; nonlinear weight functions improve only risk metrics (lower MDD, higher Calmar) without enhancing returns. This finding reinforces the robustness of the baseline: within the 427 configurations tested, 12/VIX sits at the intersection of maximum return extraction and minimum complexity, and no GARCH-based volatility forecast—despite superior prediction accuracy—translates its forecasting advantage into allocation advantage.

4.3 Insurance Pricing Reconsidered: What the Premium Actually Buys

Our international results sharpen—and partly retract—the “insurance pricing” interpretation of VT (Lai, 2026a). Across 13 markets, the overlay extracts a consistent $\sim 4\%$ /year Sharpe drag (the “premium”) and delivers broad raw MDD reduction (the apparent “payout”). The exposure-matched decomposition of Section 3.6, however, shows that most of this payout is delivered by holding roughly half the volatility of buy-and-hold on average, not by timing: a constant-weight position with the same average exposure collects nearly the same raw drawdown reduction at similar cost. The insurance analogy therefore needs restating. What the premium buys, in the international panel, is a mechanically smaller risk budget whose size co-moves with VIX—genuine conditional risk reduction, but replicable by any investor who can scale exposure directly. The breadth of the raw result—13 of 13 markets across developed and emerging economies—remains consistent with the global financial cycle framework of Miranda-Agrippino and Rey (2020), in the limited sense that a single US signal successfully proxies for global risk conditions; it is not evidence of same-risk drawdown protection.

The cross-sectional correlation between VIX sensitivity and raw Δ MDD ($r = -0.817$) should be reinterpreted in the same way. Markets more sensitive to US-driven risk-off

episodes are de-levered more aggressively by the overlay, so the association primarily prices the size of the exposure reduction rather than a protection payout. We also do not interpret VIX here as a pure expected-volatility state variable: VIX embeds the variance risk premium (Bollerslev et al., 2009) and investor demand for downside protection (Bondarenko and Bernardo, 2019). As a cost description for the single-asset US case, the insurance arithmetic remains informative: the breakeven VIX level—above which VT’s cash drag is more than offset by crisis-period savings—is approximately 26.6 for SPY. Below this threshold, VT underperforms buy-and-hold (the “premium” regime); above it, VT outperforms (the “payout” regime). In the low-VIX regime ($VIX < 15$), VT costs roughly 4.0%/year in foregone Sharpe, while in the crisis regime ($VIX > 30$), it delivers approximately 12%/year of excess protection.

The two-asset 50/50 SPY/GLD allocation also resists optimization. We test dynamic allocation schemes—regime-switching, DCC-based, and VIX-conditional tilting—against the static 50/50 benchmark. All dynamic alternatives fail the Harvey threshold ($t < 3.0$), consistent with the DCC analysis of Lai (2026a) showing that the two-asset blend is already near-optimal. This parsimony is a feature: within the specifications tested, VT’s economic value derives from two robust components—the 12/VIX scaling rule and the 50/50 equity/gold allocation—neither of which benefits from the added complexity we examined.

4.4 Limitations

Several limitations should be noted. First, our sample of 22 assets, while spanning multiple asset classes, is smaller than the 50 futures contracts studied by Hood and Raughtigan (2025). Expanding to additional commodities, fixed-income instruments, and currencies would strengthen the cross-sectional evidence. Second, our out-of-sample period (2023–2026) does not include a major financial crisis comparable to 2008, limiting our ability to test MDD protection under extreme stress. Third, VIX availability constrains the analysis to the post-1990 period for US assets and post-2007 for international ETFs. Fourth, the GJR-GARCH γ estimation requires approximately 2,000 observations (8 years), limiting

the method’s applicability to newly listed securities.

Fifth, the cross-sectional regression linking γ to TSMOM loading ($N = 22$) faces an inherent endogeneity concern: both variables are derived from the same underlying return process, creating a mechanical channel that our split-sample robustness check (Section 3.2) mitigates but does not fully resolve. A formal instrumental variable approach would provide stronger causal identification, but suitable instruments are difficult to construct in a mixed-asset cross-section of this size. Future work with a larger cross-section of futures contracts—as in [Hood and Raughtigan \(2025\)](#)’s 50-asset sample—could employ a more diverse set of instruments or within-asset-class variation to address this limitation.

Sixth, the international equity analysis (13 markets in Table 7) is now stated in exposure-matched terms: the raw drawdown improvements are attributable primarily to de-levering, and the exposure-matched gaps do not reject a no-timing null in either sample. A remaining limitation is sample-driven: the common window (post-2012) excludes the 2008–2009 crisis, while the inception-aware sample contains only that single crisis episode. The inception-aware exposure-matched gap (+5.0 pp, positive in 12 of 13 markets, $p = 0.21$, 0/13 after Holm) is therefore neither confirmed nor ruled out as a modest crisis-window timing effect; one crisis is not a test.

5 Conclusion

This paper demonstrates that VIX-based de-risking embeds a mechanical trend-following overlay through the leverage effect, but this is not the whole story. VT possesses a dual mechanism: the Sharpe ratio channel operates partially through implicit asset-specific TSMOM exposure, while the maximum drawdown channel—VT’s primary economic value—is largely independent of trend following. Across all 22 primary assets plus the 50/50 composite, block bootstrap (10,000 replications) finds that 22 of 23 rows have positive CI lower bounds (90% level), with point estimates at or above 83.8% and medians above 100% for 21 of 23 rows. The canonical five-asset results (SPY 103.7%, 50/50 95.6%, DIA 106.2%, QQQ 109.0%, IWM 102.2%) are confirmed and generalized:

TSMOM removal preserves the drawdown channel across asset classes, not just US equities, while some $> 100\%$ estimates can arise mechanically during rebound phases of momentum crashes. The leverage effect parameter (γ) predicts VT’s trend-following loading cross-sectionally ($r = 0.564$, $p = 0.006$, $N = 22$), an empirical regularity consistent with the leverage effect mechanism; a split-sample robustness check ($r = 0.793$, $p < 0.001$, split-sample canonical) survives temporal separation of the estimation windows but is partly shaped by a 2017–2026 regime shift in which international and safe-haven loadings flipped positive. We note that this cross-sectional association ($N = 22$, mixed asset classes) does not establish a causal link.

Four practical implications emerge. First, VT investors already hold embedded asset-specific trend-following exposure as a mechanical byproduct of the leverage effect. Second, a single US VIX signal reduces realized risk and raw drawdowns across global equity allocations (13/13 markets with positive raw MDD reduction; inception-aware average 27.5 pp, common-sample average 12.6 pp), but this benefit is de-levering rather than same-risk protection: a constant-weight position with the same average exposure replicates most of it, and the exposure-matched gap does not reject a no-timing null, so leverage-flexible investors should not read the raw drawdown reduction as timing skill. Third, within-equity diversification (sector rotation) does not require sector-specific VT tuning, as the gamma mechanism operates only at the cross-asset-class level. Fourth, within the specifications tested, the 12/VIX formula and the 50/50 equity/gold allocation are both highly robust: no optimization—dynamic allocation, alternative thresholds, or higher rebalancing frequency—survives the [Harvey et al. \(2016\)](#) threshold, confirming that VT’s value resides in its parsimonious risk-scaling mechanism rather than in parameter tuning.

Our findings reframe the “Is VT just trend following?” debate. VT *contains* trend following, but VT is not *reducible to* trend following. A direct test shows that five canonical trend-following strategies all fail to replicate VT’s benefits. Its economic value resides in a VIX-level channel that provides drawdown insurance—a benefit that pure momentum strategies do not replicate.

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